

My Magic UW – The Art of the Break

Badger BuildFest 2026 Live stress test on 27 September 2026 github.com/benverhaalen/magic-uw

Our agent

My Magic UW is a desktop study app for UW–Madison students. Its question agent answers questions about a student's courses. The model is the student's own Claude Code or Codex, with its tools off and a strict JSON output schema. The answer is a list of short sentences, and each sentence must cite an exact quote from course passages that code retrieved. Our rule is "AI writes, code decides." The model writes the language, and deterministic code checks every fact before the student sees it.

The failure

Our verifier checked the evidence but not the claim. The function `checkAnswer` confirmed that each cited quote was really in the course material. It did not compare the sentence with its quote. Thus a wrong date, number or name passed the check when its quote was real, and the working citation made the wrong fact look more trustworthy.

Our agent wrote this type of sentence in the live test. For "How much is the final project worth?" it cited the real quote "final project 45%" in a sentence that started "The syllabus originally set the final project at 30%." The old check accepted it.

The same pattern was in three more checks. Any email with a course code in its subject got the label "Course staff". A Jev judgment (a small hosted classifier) could raise any email to urgent. A link that named the course number became the course site, also when a student posted it in a discussion.

How we stress-tested it

We built an evaluation harness around the real agent, with 20 synthetic course scenarios that make the failure likely. They include changed syllabus dates, rooms and weights, numbers near each other, arithmetic, weekdays, and two courses on one page. Claude Code (`claude-sonnet-5`) answered each scenario 3 times through the same prompt, schema and client path as the app. We ran the harness on the first fix and on the final fix. Code scored each sentence, and we read all 167 sentences ourselves. For each of the four weak checks, we also generated 60 targeted cases with a scripted fake model. These cases always contain the trigger, so they measure the fix and not the natural rate.

How often it happened

On the real agent, 1 sentence in 167 (0.6%) contradicted its own cited quote. The rate is low, but the old check showed every such sentence with a real citation. Before the fix, each weak check failed in every targeted case.

Live test on our agent, Wilson 95% intervals in brackets	First fix	Final fix
Sentences that contradict their own cited quote	0 of 87	1 of 80 [0.2, 6.7]%, caught
Contradicting sentences that the student sees	0 of 87	0 of 77 [0, 4.8]%
Answers where the check replaced a correct sentence	15 of 60 [15.8, 37.2]%	2 of 60 [0.9, 11.4]%
Targeted cases, 60 for each weak check	Before	After
Wrong sentence shown next to a real quote	60 of 60	0 of 60
Each of the three mail and site checks	60 of 60	0 of 60
Correct results kept (staff mail urgent, real sites found)	60 of 60	60 of 60

What we did

Our first fix (`0f710ed`) makes the verifier check the claim too. Every date, weekday, number and name in a sentence must now appear in its cited quote (`claimsMatch`). If one does not, the app shows the quote word for word in place of the sentence. This fix passed every synthetic test, but on the live agent it rejected correct sentences in 15 of 60 answers, mostly calculated numbers.

Our final fix (`459669c`) gives calculated numbers their own path. The prompt tells the agent to write the arithmetic from quoted numbers, for example "8 – 1 = 7, so 7 × 25 = 175 points." Code then does the arithmetic again. False rejects fell to 2 of 60 answers. In the other three checks, only known staff addresses can make mail urgent, a Jev judgment can raise mail by one level at most, and a link only in a discussion never becomes the course site.

What we learned

- A verifier must check the claim, not only the evidence. A real citation on a wrong sentence is worse than no citation.
- Failures that we write ourselves do not predict real model output. Our fix passed every synthetic test and then rejected correct sentences in 1 of 4 live answers. Measure every fix on the real agent.
- Let the model show its work, and let code check it again.
- When code confirms something, ask what it really proved.

All scenarios are synthetic, with no student data. See the [live harness](#), raw answers for the [final fix](#) and the [first fix](#), the [targeted cases](#), and the [full write-up](#).